Dataset	Cluster	Phase	Matched GO Term Count	Weighted Score (-log10 adj p)	Matched GO Terms	Matched Subcategories	Matched Keywords	Matched Genes	Matched GO Terms with Genes	Matched Terms with Weights
acute	1	Proliferation / tissue formation	7	52.43512553	extracellular matrix organization (GO:0030198); extracellular structure organization (GO:0043062); external encapsulating structure organization (GO:0045229); collagen fibril organization (GO:0030199); eye morphogenesis (GO:0048592); muscle organ development (GO:0007517); supramolecular fiber organization (GO:0097435)	ECM deposition / proliferation; ECM deposition / proliferation; ECM deposition / proliferation; Proliferation / tissue formation; Proliferation / tissue formation; ECM deposition / proliferation	extracellular matrix; extracellular structure organization; external encapsulating structure organization; collagen fibril organization; morphogenesis; development; supramolecular fiber organization	FBN2;LAMA2;COL14A1;COL12A1;PLOD2;ITGAE;ADAMTS12;NID2;COL19A1;ADAMTS15;ADAMTS6;TGFB2;VWF;CMA1;FN1;BGN;P3H1;MMP12;MFAP5;COL1A2;LOX;COL5A1;P4HA2;P4HA3;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;FBN1; FBN2;VWF;LAMA2;COL14A1;FN1;BGN;ITGAE;ADAMTS12;NID2;COL19A1;MMP12;MFAP5;ADAMTS15;COL1A2;LOX;COL5A1;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;ADAMTS6;FBN1; FBN2;VWF;LAMA2;COL14A1;FN1;BGN;ITGAE;ADAMTS12;NID2;COL19A1;MMP12;MFAP5;ADAMTS15;COL1A2;LOX;COL5A1;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;ADAMTS6;FBN1; TGFB2;COL14A1;COL12A1;P3H1;PLOD2;COL19A1;COL1A2;LOX;COL5A1;P4HA2;P4HA3;PXDN;COL5A2;COL6A3;COL8A1; FBN2;MFAP5;TENM3;COL5A1;COL5A2;FBN1; SGCE;MEF2C;MYH3;LAMA2;COL6A3;ITGB1BP2;IGF1;MKX; TGFB2;COL14A1;COL12A1;P3H1;PLOD2;MYO7A;COL19A1;COL1A2;ACTC1;LOX;COL5A1;P4HA2;P4HA3;PXDN;COL5A2;COL8A1;COL6A3;EMILIN1	extracellular matrix organization (GO:0030198): FBN2;LAMA2;COL14A1;COL12A1;PLOD2;ITGAE;ADAMTS12;NID2;COL19A1;ADAMTS15;ADAMTS6;TGFB2;VWF;CMA1;FN1;BGN;P3H1;MMP12;MFAP5;COL1A2;LOX;COL5A1;P4HA2;P4HA3;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;FBN1 | extracellular structure organization (GO:0043062): FBN2;VWF;LAMA2;COL14A1;FN1;BGN;ITGAE;ADAMTS12;NID2;COL19A1;MMP12;MFAP5;ADAMTS15;COL1A2;LOX;COL5A1;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;ADAMTS6;FBN1 | external encapsulating structure organization (GO:0045229): FBN2;VWF;LAMA2;COL14A1;FN1;BGN;ITGAE;ADAMTS12;NID2;COL19A1;MMP12;MFAP5;ADAMTS15;COL1A2;LOX;COL5A1;PXDN;ADAM12;COL5A2;COL8A1;COL6A3;RECK;ADAMTS6;FBN1 | collagen fibril organization (GO:0030199): TGFB2;COL14A1;COL12A1;P3H1;PLOD2;COL19A1;COL1A2;LOX;COL5A1;P4HA2;P4HA3;PXDN;COL5A2;COL6A3;COL8A1 | eye morphogenesis (GO:0048592): FBN2;MFAP5;TENM3;COL5A1;COL5A2;FBN1 | muscle organ development (GO:0007517): SGCE;MEF2C;MYH3;LAMA2;COL6A3;ITGB1BP2;IGF1;MKX | supramolecular fiber organization (GO:0097435): TGFB2;COL14A1;COL12A1;P3H1;PLOD2;MYO7A;COL19A1;COL1A2;ACTC1;LOX;COL5A1;P4HA2;P4HA3;PXDN;COL5A2;COL8A1;COL6A3;EMILIN1	extracellular matrix organization (GO:0030198) (ECM deposition / proliferation: extracellular matrix, weight=12.942); extracellular structure organization (GO:0043062) (ECM deposition / proliferation: extracellular structure organization, weight=10.521); external encapsulating structure organization (GO:0045229) (ECM deposition / proliferation: external encapsulating structure organization, weight=10.521); collagen fibril organization (GO:0030199) (ECM deposition / proliferation: collagen fibril organization, weight=8.538); eye morphogenesis (GO:0048592) (Proliferation / tissue formation: morphogenesis, weight=3.858); muscle organ development (GO:0007517) (Proliferation / tissue formation: development, weight=3.268); supramolecular fiber organization (GO:0097435) (ECM deposition / proliferation: supramolecular fiber organization, weight=2.787)
acute	1	ECM / remodeling	4	13.64336138	actin-myosin filament sliding (GO:0033275); muscle filament sliding (GO:0030049); striated muscle contraction (GO:0006941); muscle organ development (GO:0007517)	Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling	actin-myosin; muscle; muscle; muscle	MYL4;ACTC1;MYH3;TNNT1;TNNC1;TNNT2;TNNI1; MYL4;ACTC1;MYH3;TNNT1;TNNC1;TNNT2;TNNI1; MYL4;CHRNA1;MYH3;TNNT1;TNNC1;TNNT2;CASQ2;TNNI1; SGCE;MEF2C;MYH3;LAMA2;COL6A3;ITGB1BP2;IGF1;MKX	actin-myosin filament sliding (GO:0033275): MYL4;ACTC1;MYH3;TNNT1;TNNC1;TNNT2;TNNI1 | muscle filament sliding (GO:0030049): MYL4;ACTC1;MYH3;TNNT1;TNNC1;TNNT2;TNNI1 | striated muscle contraction (GO:0006941): MYL4;CHRNA1;MYH3;TNNT1;TNNC1;TNNT2;CASQ2;TNNI1 | muscle organ development (GO:0007517): SGCE;MEF2C;MYH3;LAMA2;COL6A3;ITGB1BP2;IGF1;MKX	actin-myosin filament sliding (GO:0033275) (Muscle / contractile remodeling: actin-myosin, weight=3.459); muscle filament sliding (GO:0030049) (Muscle / contractile remodeling: muscle, weight=3.459); striated muscle contraction (GO:0006941) (Muscle / contractile remodeling: muscle, weight=3.459); muscle organ development (GO:0007517) (Muscle / contractile remodeling: muscle, weight=3.268)
acute	1	Inflammation	0	0						
acute	1	Stress / damage response	0	0						
acute	1	Cellular regulation / turnover	0	0						
acute	1	Ambiguous / specialized signaling	0	0						
acute	2	Ambiguous / specialized signaling	1	1.431789342	activation of adenylate cyclase activity (GO:0007190)	Ambiguous signaling	adenylate cyclase	ADORA2B;ADCY1	activation of adenylate cyclase activity (GO:0007190): ADORA2B;ADCY1	activation of adenylate cyclase activity (GO:0007190) (Ambiguous signaling: adenylate cyclase, weight=1.432)
acute	2	Inflammation	0	0						
acute	2	Proliferation / tissue formation	0	0						
acute	2	ECM / remodeling	0	0						
acute	2	Stress / damage response	0	0						
acute	2	Cellular regulation / turnover	0	0						
acute	3	Ambiguous / specialized signaling	1	2.248148429	positive regulation of cation channel activity (GO:2001259)	Ambiguous signaling	cation channel	P2RY6;GAL;CTSS;CCR2	positive regulation of cation channel activity (GO:2001259): P2RY6;GAL;CTSS;CCR2	positive regulation of cation channel activity (GO:2001259) (Ambiguous signaling: cation channel, weight=2.248)
acute	3	Inflammation	1	1.353259478	negative regulation of myeloid leukocyte mediated immunity (GO:0002887)	Inflammation	leukocyte	CD300A;CCR2	negative regulation of myeloid leukocyte mediated immunity (GO:0002887): CD300A;CCR2	negative regulation of myeloid leukocyte mediated immunity (GO:0002887) (Inflammation: leukocyte, weight=1.353)
acute	3	Proliferation / tissue formation	0	0						
acute	3	ECM / remodeling	0	0						
acute	3	Stress / damage response	0	0						
acute	3	Cellular regulation / turnover	0	0						
acute	4	Inflammation	7	10.12935783	positive regulation of immune response (GO:0050778); lymphocyte chemotaxis (GO:0048247); positive regulation of response to biotic stimulus (GO:0002833); mast cell mediated immunity (GO:0002448); regulation of T cell activation (GO:0050863); inflammatory response (GO:0006954); positive regulation of interferon-gamma production (GO:0032729)	Inflammation; Inflammation; Inflammation; Inflammation; Inflammation; Inflammation; Inflammation	immune; lymphocyte; biotic stimulus; t cell; t cell; inflammatory; interferon-gamma	SASH3;CCL5;TLR8;CYBA; CCL5;CYP7B1;PIK3CG; CCL5;TLR8;CYBA; SERPINB9;PIK3CG; CCL5;PTPN22;TNFAIP8L2; PTGFR;CCL5;C3AR1;CYBA;PIK3CG; EBI3;TLR8;PTPN22	positive regulation of immune response (GO:0050778): SASH3;CCL5;TLR8;CYBA | lymphocyte chemotaxis (GO:0048247): CCL5;CYP7B1;PIK3CG | positive regulation of response to biotic stimulus (GO:0002833): CCL5;TLR8;CYBA | mast cell mediated immunity (GO:0002448): SERPINB9;PIK3CG | regulation of T cell activation (GO:0050863): CCL5;PTPN22;TNFAIP8L2 | inflammatory response (GO:0006954): PTGFR;CCL5;C3AR1;CYBA;PIK3CG | positive regulation of interferon-gamma production (GO:0032729): EBI3;TLR8;PTPN22	positive regulation of immune response (GO:0050778) (Inflammation: immune, weight=1.509); lymphocyte chemotaxis (GO:0048247) (Inflammation: lymphocyte, weight=1.464); positive regulation of response to biotic stimulus (GO:0002833) (Inflammation: biotic stimulus, weight=1.464); mast cell mediated immunity (GO:0002448) (Inflammation: t cell, weight=1.464); regulation of T cell activation (GO:0050863) (Inflammation: t cell, weight=1.464); inflammatory response (GO:0006954) (Inflammation: inflammatory, weight=1.401); positive regulation of interferon-gamma production (GO:0032729) (Inflammation: interferon-gamma, weight=1.362)
acute	4	Proliferation / tissue formation	0	0						
acute	4	ECM / remodeling	0	0						
acute	4	Stress / damage response	0	0						
acute	4	Cellular regulation / turnover	0	0						
acute	4	Ambiguous / specialized signaling	0	0						
acute	5	Ambiguous / specialized signaling	2	5.716191205	generation of neurons (GO:0048699); neuron differentiation (GO:0030182)	Neural repair / ambiguous; Neural repair / ambiguous	neuron; neuron	HEYL;SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;MDGA1;MARK1; SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;GLI2	generation of neurons (GO:0048699): HEYL;SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;MDGA1;MARK1 | neuron differentiation (GO:0030182): SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;GLI2	generation of neurons (GO:0048699) (Neural repair / ambiguous: neuron, weight=3.417); neuron differentiation (GO:0030182) (Neural repair / ambiguous: neuron, weight=2.299)
acute	5	Proliferation / tissue formation	1	2.299288184	neuron differentiation (GO:0030182)	Proliferation / tissue formation	differentiation	SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;GLI2	neuron differentiation (GO:0030182): SHH;FZD2;LHX2;WNT5A;DDIT4;IRX4;WNT9A;GLI2	neuron differentiation (GO:0030182) (Proliferation / tissue formation: differentiation, weight=2.299)
acute	5	Inflammation	0	0						
acute	5	ECM / remodeling	0	0						
acute	5	Stress / damage response	0	0						
acute	5	Cellular regulation / turnover	0	0						
acute	6	Proliferation / tissue formation	10	108.4742762	microtubule cytoskeleton organization involved in mitosis (GO:1902850); mitotic spindle organization (GO:0007052); mitotic sister chromatid segregation (GO:0000070); mitotic nuclear division (GO:0140014); sister chromatid segregation (GO:0000819); positive regulation of mitotic cell cycle phase transition (GO:1901992); regulation of cell cycle process (GO:0010564); cytoskeleton-dependent cytokinesis (GO:0061640); regulation of mitotic metaphase/anaphase transition (GO:0030071); mitotic chromosome condensation (GO:0007076)	Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation; Cell cycle / proliferation	mitosis; mitotic; mitotic; mitotic; sister chromatid segregation; cell cycle; cell cycle; cytokinesis; mitotic; mitotic	STIL;ZWILCH;PLK1;BUB1B;KIF11;SKA1;AURKB;SGO1;CENPE;CCNB1;CENPF;ESPL1;CENPH;CENPI;NUF2;NUSAP1;BIRC5;BUB1; STIL;ZWILCH;PLK1;BUB1B;KIF23;KIF11;SKA1;AURKB;SGO1;CENPE;CCNB1;CENPF;CENPH;KIFC1;CENPI;NUF2;BIRC5;BUB1; SPAG5;CDCA5;PLK1;NCAPG;KIF23;KIF22;SGO1;CENPE;CCNB1;KIF18B;ESPL1;KIFC1;NUSAP1;CDK1;CEP55; SGO1;KIF18B;ESPL1;SPAG5;KIFC1;UBE2C;PLK1;NUSAP1;KIF11; SGO1;KIF18B;SPAG5;ESPL1;KIFC1;PLK1;NUSAP1; CCNB1;ESPL1;UBE2C;CDCA5;CDK1;FBXO5;CDC25C;DTL; SPAG5;PLK1;KIF11;FBXO5;MKI67;CDC25C;ECT2;BUB1;AURKB; ESPL1;PLK1;NUSAP1;CKAP2;KIF23;ECT2;CEP55;AURKB; CENPE;CCNB1;ESPL1;UBE2C;PLK1;FBXO5; CCNB1;CDCA5;PLK1;NUSAP1;CDK1;NCAPG	microtubule cytoskeleton organization involved in mitosis (GO:1902850): STIL;ZWILCH;PLK1;BUB1B;KIF11;SKA1;AURKB;SGO1;CENPE;CCNB1;CENPF;ESPL1;CENPH;CENPI;NUF2;NUSAP1;BIRC5;BUB1 | mitotic spindle organization (GO:0007052): STIL;ZWILCH;PLK1;BUB1B;KIF23;KIF11;SKA1;AURKB;SGO1;CENPE;CCNB1;CENPF;CENPH;KIFC1;CENPI;NUF2;BIRC5;BUB1 | mitotic sister chromatid segregation (GO:0000070): SPAG5;CDCA5;PLK1;NCAPG;KIF23;KIF22;SGO1;CENPE;CCNB1;KIF18B;ESPL1;KIFC1;NUSAP1;CDK1;CEP55 | mitotic nuclear division (GO:0140014): SGO1;KIF18B;ESPL1;SPAG5;KIFC1;UBE2C;PLK1;NUSAP1;KIF11 | sister chromatid segregation (GO:0000819): SGO1;KIF18B;SPAG5;ESPL1;KIFC1;PLK1;NUSAP1 | positive regulation of mitotic cell cycle phase transition (GO:1901992): CCNB1;ESPL1;UBE2C;CDCA5;CDK1;FBXO5;CDC25C;DTL | regulation of cell cycle process (GO:0010564): SPAG5;PLK1;KIF11;FBXO5;MKI67;CDC25C;ECT2;BUB1;AURKB | cytoskeleton-dependent cytokinesis (GO:0061640): ESPL1;PLK1;NUSAP1;CKAP2;KIF23;ECT2;CEP55;AURKB | regulation of mitotic metaphase/anaphase transition (GO:0030071): CENPE;CCNB1;ESPL1;UBE2C;PLK1;FBXO5 | mitotic chromosome condensation (GO:0007076): CCNB1;CDCA5;PLK1;NUSAP1;CDK1;NCAPG	microtubule cytoskeleton organization involved in mitosis (GO:1902850) (Cell cycle / proliferation: mitosis, weight=19.679); mitotic spindle organization (GO:0007052) (Cell cycle / proliferation: mitotic, weight=18.317); mitotic sister chromatid segregation (GO:0000070) (Cell cycle / proliferation: mitotic, weight=16.703); mitotic nuclear division (GO:0140014) (Cell cycle / proliferation: mitotic, weight=8.548); sister chromatid segregation (GO:0000819) (Cell cycle / proliferation: sister chromatid segregation, weight=8.034); positive regulation of mitotic cell cycle phase transition (GO:1901992) (Cell cycle / proliferation: cell cycle, weight=8.026); regulation of cell cycle process (GO:0010564) (Cell cycle / proliferation: cell cycle, weight=7.375); cytoskeleton-dependent cytokinesis (GO:0061640) (Cell cycle / proliferation: cytokinesis, weight=7.375); regulation of mitotic metaphase/anaphase transition (GO:0030071) (Cell cycle / proliferation: mitotic, weight=7.222); mitotic chromosome condensation (GO:0007076) (Cell cycle / proliferation: mitotic, weight=7.195)
acute	6	Inflammation	1	7.375181874	cytoskeleton-dependent cytokinesis (GO:0061640)	Inflammation	cytokine	ESPL1;PLK1;NUSAP1;CKAP2;KIF23;ECT2;CEP55;AURKB	cytoskeleton-dependent cytokinesis (GO:0061640): ESPL1;PLK1;NUSAP1;CKAP2;KIF23;ECT2;CEP55;AURKB	cytoskeleton-dependent cytokinesis (GO:0061640) (Inflammation: cytokine, weight=7.375)
acute	6	ECM / remodeling	0	0						
acute	6	Stress / damage response	0	0						
acute	6	Cellular regulation / turnover	0	0						
acute	6	Ambiguous / specialized signaling	0	0						
acute	7	ECM / remodeling	8	26.9684777	muscle contraction (GO:0006936); heart contraction (GO:0060047); myofibril assembly (GO:0030239); striated muscle cell development (GO:0055002); striated muscle contraction (GO:0006941); actomyosin structure organization (GO:0031032); cardiac muscle contraction (GO:0060048); skeletal muscle tissue development (GO:0007519)	Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling	muscle; contraction; myofibril; muscle; muscle; actomyosin; muscle; muscle	RYR1;TMOD1;CKMT2;MYLK2;MYBPC2;TMOD4;MYOT;PGAM2;TRDN; MYLK2;ATP1A2;SCN4B;TRDN; TMOD1;OBSCN;TMOD4;MYOZ1; TMOD1;TMOD4;MYOZ1; MYLK2;PGAM2;ATP1A2;SCN4B; TMOD1;OBSCN;TMOD4;MYOZ1; MYLK2;ATP1A2;SCN4B; RYR1;MYLK2;MYOZ1	muscle contraction (GO:0006936): RYR1;TMOD1;CKMT2;MYLK2;MYBPC2;TMOD4;MYOT;PGAM2;TRDN | heart contraction (GO:0060047): MYLK2;ATP1A2;SCN4B;TRDN | myofibril assembly (GO:0030239): TMOD1;OBSCN;TMOD4;MYOZ1 | striated muscle cell development (GO:0055002): TMOD1;TMOD4;MYOZ1 | striated muscle contraction (GO:0006941): MYLK2;PGAM2;ATP1A2;SCN4B | actomyosin structure organization (GO:0031032): TMOD1;OBSCN;TMOD4;MYOZ1 | cardiac muscle contraction (GO:0060048): MYLK2;ATP1A2;SCN4B | skeletal muscle tissue development (GO:0007519): RYR1;MYLK2;MYOZ1	muscle contraction (GO:0006936) (Muscle / contractile remodeling: muscle, weight=7.674); heart contraction (GO:0060047) (Muscle / contractile remodeling: contraction, weight=3.131); myofibril assembly (GO:0030239) (Muscle / contractile remodeling: myofibril, weight=3.131); striated muscle cell development (GO:0055002) (Muscle / contractile remodeling: muscle, weight=3.131); striated muscle contraction (GO:0006941) (Muscle / contractile remodeling: muscle, weight=2.933); actomyosin structure organization (GO:0031032) (Muscle / contractile remodeling: actomyosin, weight=2.489); cardiac muscle contraction (GO:0060048) (Muscle / contractile remodeling: muscle, weight=2.261); skeletal muscle tissue development (GO:0007519) (Muscle / contractile remodeling: muscle, weight=2.218)
acute	7	Proliferation / tissue formation	2	5.349515802	striated muscle cell development (GO:0055002); skeletal muscle tissue development (GO:0007519)	Proliferation / tissue formation; Proliferation / tissue formation	development; development	TMOD1;TMOD4;MYOZ1; RYR1;MYLK2;MYOZ1	striated muscle cell development (GO:0055002): TMOD1;TMOD4;MYOZ1 | skeletal muscle tissue development (GO:0007519): RYR1;MYLK2;MYOZ1	striated muscle cell development (GO:0055002) (Proliferation / tissue formation: development, weight=3.131); skeletal muscle tissue development (GO:0007519) (Proliferation / tissue formation: development, weight=2.218)
acute	7	Ambiguous / specialized signaling	2	4.887992799	regulation of release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0010880); release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0014808)	Ambiguous signaling; Ambiguous signaling	calcium; calcium	DHRS7C;ATP1A2;TRDN; RYR1;TRDN	regulation of release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0010880): DHRS7C;ATP1A2;TRDN | release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0014808): RYR1;TRDN	regulation of release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0010880) (Ambiguous signaling: calcium, weight=2.627); release of sequestered calcium ion into cytosol by sarcoplasmic reticulum (GO:0014808) (Ambiguous signaling: calcium, weight=2.261)
acute	7	Inflammation	0	0						
acute	7	Stress / damage response	0	0						
acute	7	Cellular regulation / turnover	0	0						
acute	9	Proliferation / tissue formation	4	10.09683892	regulation of muscle cell differentiation (GO:0051147); regulation of skeletal muscle fiber development (GO:0048742); regulation of myotube differentiation (GO:0010830); positive regulation of muscle cell differentiation (GO:0051149)	Proliferation / tissue formation; Proliferation / tissue formation; Proliferation / tissue formation; Proliferation / tissue formation	differentiation; development; differentiation; differentiation	MYOG;CDH15;HDAC9; MYOG;HDAC9; MYOG;HDAC9; MYOG;CDH15	regulation of muscle cell differentiation (GO:0051147): MYOG;CDH15;HDAC9 | regulation of skeletal muscle fiber development (GO:0048742): MYOG;HDAC9 | regulation of myotube differentiation (GO:0010830): MYOG;HDAC9 | positive regulation of muscle cell differentiation (GO:0051149): MYOG;CDH15	regulation of muscle cell differentiation (GO:0051147) (Proliferation / tissue formation: differentiation, weight=3.289); regulation of skeletal muscle fiber development (GO:0048742) (Proliferation / tissue formation: development, weight=2.798); regulation of myotube differentiation (GO:0010830) (Proliferation / tissue formation: differentiation, weight=2.005); positive regulation of muscle cell differentiation (GO:0051149) (Proliferation / tissue formation: differentiation, weight=2.005)
acute	9	ECM / remodeling	3	8.091930391	regulation of muscle cell differentiation (GO:0051147); regulation of skeletal muscle fiber development (GO:0048742); positive regulation of muscle cell differentiation (GO:0051149)	Muscle / contractile remodeling; Muscle / contractile remodeling; Muscle / contractile remodeling	muscle; muscle; muscle	MYOG;CDH15;HDAC9; MYOG;HDAC9; MYOG;CDH15	regulation of muscle cell differentiation (GO:0051147): MYOG;CDH15;HDAC9 | regulation of skeletal muscle fiber development (GO:0048742): MYOG;HDAC9 | positive regulation of muscle cell differentiation (GO:0051149): MYOG;CDH15	regulation of muscle cell differentiation (GO:0051147) (Muscle / contractile remodeling: muscle, weight=3.289); regulation of skeletal muscle fiber development (GO:0048742) (Muscle / contractile remodeling: muscle, weight=2.798); positive regulation of muscle cell differentiation (GO:0051149) (Muscle / contractile remodeling: muscle, weight=2.005)
acute	9	Inflammation	0	0						
acute	9	Stress / damage response	0	0						
acute	9	Cellular regulation / turnover	0	0						
acute	9	Ambiguous / specialized signaling	0	0						
acute	12	Proliferation / tissue formation	6	9.960639396	aortic valve morphogenesis (GO:0003180); aortic valve development (GO:0003176); metanephros development (GO:0001656); negative regulation of supramolecular fiber organization (GO:1902904); negative regulation of microtubule polymerization or depolymerization (GO:0031111); cell migration involved in heart development (GO:0060973)	Proliferation / tissue formation; Proliferation / tissue formation; Proliferation / tissue formation; ECM deposition / proliferation; Cell cycle / proliferation; Proliferation / tissue formation	morphogenesis; development; development; supramolecular fiber organization; microtubule polymerization; development	ROBO2;ELN;HEY2;TWIST1; ROBO2;ELN;HEY2;TWIST1; ROBO2;OSR2;FOXD1; CLIP3;STMN2;FGF13;KANK4; CLIP3;STMN2;FGF13; PDGFRB;PITX2	aortic valve morphogenesis (GO:0003180): ROBO2;ELN;HEY2;TWIST1 | aortic valve development (GO:0003176): ROBO2;ELN;HEY2;TWIST1 | metanephros development (GO:0001656): ROBO2;OSR2;FOXD1 | negative regulation of supramolecular fiber organization (GO:1902904): CLIP3;STMN2;FGF13;KANK4 | negative regulation of microtubule polymerization or depolymerization (GO:0031111): CLIP3;STMN2;FGF13 | cell migration involved in heart development (GO:0060973): PDGFRB;PITX2	aortic valve morphogenesis (GO:0003180) (Proliferation / tissue formation: morphogenesis, weight=2.212); aortic valve development (GO:0003176) (Proliferation / tissue formation: development, weight=2.212); metanephros development (GO:0001656) (Proliferation / tissue formation: development, weight=1.432); negative regulation of supramolecular fiber organization (GO:1902904) (ECM deposition / proliferation: supramolecular fiber organization, weight=1.420); negative regulation of microtubule polymerization or depolymerization (GO:0031111) (Cell cycle / proliferation: microtubule polymerization, weight=1.385); cell migration involved in heart development (GO:0060973) (Proliferation / tissue formation: development, weight=1.301)
acute	12	Cellular regulation / turnover	1	1.431520285	negative regulation of protein polymerization (GO:0032272)	Protein turnover / cellular regulation	protein polymerization	CLIP3;STMN2;KANK4	negative regulation of protein polymerization (GO:0032272): CLIP3;STMN2;KANK4	negative regulation of protein polymerization (GO:0032272) (Protein turnover / cellular regulation: protein polymerization, weight=1.432)
acute	12	Inflammation	0	0						
acute	12	ECM / remodeling	0	0						
acute	12	Stress / damage response	0	0						
acute	12	Ambiguous / specialized signaling	0	0						

acute	13	Inflammation	9	114.1095048	cytokine-mediated signaling pathway (GO:0019221); cellular response to cytokine stimulus (GO:0071345); defense response to symbiont (GO:0140546); positive regulation of tumor necrosis factor superfamily cytokine production (GO:1903557); defense response to virus (GO:0051607); regulation of tumor necrosis factor production (GO:0032680); positive regulation of tumor necrosis factor production (GO:0032760); negative regulation of cytokine production (GO:0001818); regulation of inflammatory response (GO:0050727)	Inflammation; Inflammation; Inflammation; Inflammation; Inflammation; Inflammation; Inflammation; Inflammation	cytokine; cytokine; defense response; cytokine; defense response; tumor necrosis factor; tumor necrosis factor; cytokine; inflammatory	CD86;IFITM1;IL1RN;CD40;SPI1;CSF3R;ITGAM;CD80;ITGB2;PTPRJ;IL1RAP;TNF;IL18BP;LBP;CCR5;IL10;IFNAR2;FCER1G;SERPINB2;SYK;IL15;IL1R2;TRAF1;TNFRSF1B;VAV1;HCK;OAS2;IRF7;LCP1;RNASEL;SP100;CEBPD;CSF2RB;SAMHD1;IL2RG;CSF2RA;LRP8;SOCS3;BCL2L11;STAT4;GBP2;IL33;STAT1;IFNB1;IL10RA;MX2;PSMB8;PML;BST2;TNFSF9;XAF1;IL7R;PF4; CD86;RIPOR2;IL1RN;CD40;SELPLG;CSF3R;ITGAM;CEBPD;PDE1B;CD80;ITGB2;CXCR4;PTPRJ;CSF2RB;IL1RAP;IL2RG;TNF;CSF2RA;LRP8;IL18BP;SOCS3;SBNO2;BCL2L11;STAT4;HAS2;LGALS9L;BP;CCR5;GBP2;GBP3;IL10;IFNAR2;GBP7;SYK;STAT1;IFNB1;IL1R2;IL10RA;TNFRSF1B;VAV1;HCK;HCLS1;TLR4;TLR2;PF4; IFNAR2;ZBP1;RTP4;IFITM1;CD40;RNASEL;MLKL;RIPK3;IFNB1;STAT1;MX2;SAMHD1;IFI44L;BST2;IFIH1;PLSCR1;PTPRC;STING1;OAS2;IRF7;PNAP1;TLR3;TLR2; CD86;IL33;SYK;LY96;PTPRJ;THBS1;TIRAP;IFIH1;TYROBP;PTPRC;OAS2;LBP;LGALS9;TLR4;TLR3; TMEM106A;HAVCR2;TLR2;PF4; IFNAR2;ZBP1;RTP4;IFITM1;CD40;RNASEL;MLKL;RIPK3;IFNB1;STAT1;MX2; SAMHD1;IFI44L;BST2;IFIH1;PLSCR1;PTPRC;STING1;OAS2;IRF7;PNAP1;TLR3;TLR2; IL10;IL33;ARG2;SYK; LY96;TNFAIP3;PTPRJ;THBS1;TIRAP;IFIH1;TYROBP;PTPRC;OAS2;LGALS9;LBP;TLR4;CD33;TLR3;TMEM106A; HAVCR2;TLR2;PF4; IL33;SYK;LY96;PTPRJ;THBS1;TIRAP;IFIH1;TYROBP;PTPRC;OAS2;LBP;LGALS9;TLR4; TLR3;TMEM106A;HAVCR2;TLR2;PF4; IL10;SRGN;CD274;CD84;MACIR;UBA7;TNFAIP3;PDCD1LG2;MEFV; TNF;THBS1;IFIH1;CD200R1;TYROBP;PTPRC;DHX58;NCKAP1L;LGALS9;LGALS7;IKBKE;TLR4;CD33;HAVCR2; IL10;ZBP1;IL33;MACIR;IL15;TNFAIP3;NMI;MEFV;TNFRSF1B;PLA2G7;TNFUSP18;HCK;CD200R1;SBNO2; CASP4;NFKBIZ;PIK3AP1;TLR4;TLR3;S100A8;GPSM3;PTGES;TLR2	cytokine-mediated signaling pathway (GO:0019221): CD86;IFITM1;IL1RN;CD40;SPI1;CSF3R;ITGAM;CD80;ITGB2;PTPRJ;IL1RAP; TNF;IL18BP;LBP;CCR5;IL10;IFNAR2;FCER1G;SERPINB2;SYK;IL15;IL1R2;TRAF1;TNFRSF1B;VAV1;HCK;OAS2;IRF7;LCP1; RNASEL;SP100;CEBPD;CSF2RB;SAMHD1;IL2RG;CSF2RA;LRP8;SOCS3;BCL2L11;STAT4;GBP2;IL33;STAT1;IFNB1;IL10RA; MX2;PSMB8;PML;BST2;TNFSF9;XAF1;IL7R;PF4	cytokine-mediated signaling pathway (GO:0019221) (Inflammation: cytokine, weight=16.505); cellular response to cytokine stimulus (GO:0071345) (Inflammation: cytokine, weight=15.309); defense response to symbiont (GO:0140546) (Inflammation: defense response, weight=13.142); positive regulation of tumor necrosis factor superfamily cytokine production (GO:1903557) (Inflammation: cytokine, weight=12.662); defense response to virus (GO:0051607) (Inflammation: defense response, weight=12.662); regulation of tumor necrosis factor production (GO:0032680) (Inflammation: tumor necrosis factor, weight=12.342); positive regulation of tumor necrosis factor production (GO:0032760) (Inflammation: tumor necrosis factor, weight=12.014); negative regulation of cytokine production (GO:0001818) (Inflammation: cytokine, weight=9.847); regulation of inflammatory response (GO:0050727) (Inflammation: inflammatory, weight=9.627)
acute	13	Ambiguous / specialized signaling	1	16.50520427	cytokine-mediated signaling pathway (GO:0019221)	Ambiguous signaling	signaling	CD86;IFITM1;IL1RN;CD40;SPI1;CSF3R;ITGAM;CD80;ITGB2;PTPRJ;IL1RAP;TNF;IL18BP;LBP;CCR5;IL10; IFNAR2;FCER1G;SERPINB2;SYK;IL15;IL1R2;TRAF1;TNFRSF1B;VAV1;HCK;OAS2;IRF7;LCP1;RNASEL;SP100; CEBPD;CSF2RB;SAMHD1;IL2RG;CSF2RA;LRP8;SOCS3;BCL2L11;STAT4;GBP2;IL33;STAT1;IFNB1;IL10RA; MX2;PSMB8;PML;BST2;TNFSF9;XAF1;IL7R;PF4	cytokine-mediated signaling pathway (GO:0019221): CD86;IFITM1;IL1RN;CD40;SPI1;CSF3R;ITGAM;CD80;ITGB2;PTPRJ;IL1RAP; TNF;IL18BP;LBP;CCR5;IL10;IFNAR2;FCER1G;SERPINB2;SYK;IL15;IL1R2;TRAF1;TNFRSF1B;VAV1;HCK;OAS2;IRF7;LCP1; RNASEL;SP100;CEBPD;CSF2RB;SAMHD1;IL2RG;CSF2RA;LRP8;SOCS3;BCL2L11;STAT4;GBP2;IL33;STAT1;IFNB1;IL10RA; MX2;PSMB8;PML;BST2;TNFSF9;XAF1;IL7R;PF4	cytokine-mediated signaling pathway (GO:0019221) (Ambiguous signaling: signaling, weight=16.505)
acute	13	Stress / damage response	1	9.476577469	positive regulation of response to external stimulus (GO:0032103)	Stress / damage response	external stimulus	ZBP1;IL33;IL15;IFNB1;RARRES2;LY96;NMI;DDX60;TNF;PLA2G7;THBS1;THBD;PLSCR1;DHX58;NFKBIZ;TLR4; S100A8;GPSM3;TLR3;TLR2	positive regulation of response to external stimulus (GO:0032103): ZBP1;IL33;IL15;IFNB1;RARRES2;LY96;NMI;DDX60;TNF; PLA2G7;THBS1;THBD;PLSCR1;DHX58;NFKBIZ;TLR4;S100A8;GPSM3;TLR3;TLR2	positive regulation of response to external stimulus (GO:0032103) (Stress / damage response: external stimulus, weight=9.477)
acute	13	Proliferation / tissue formation	0	0							
acute	13	ECM / remodeling	0	0							
acute	13	Cellular regulation / turnover	0	0							